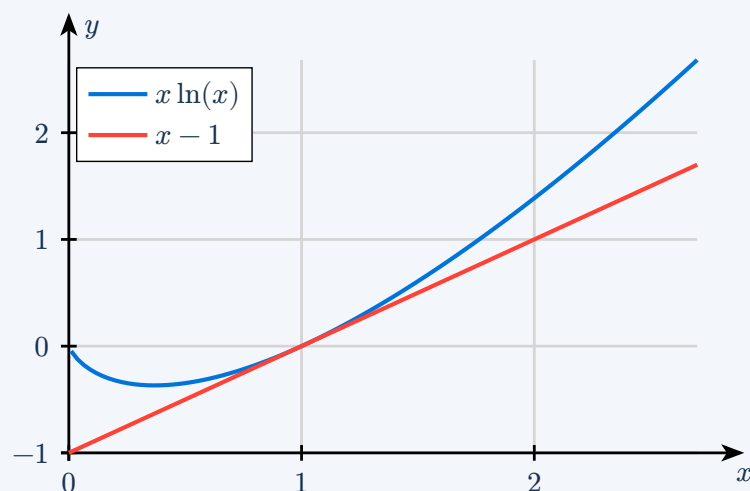


Convexity and Jensen's Inequality

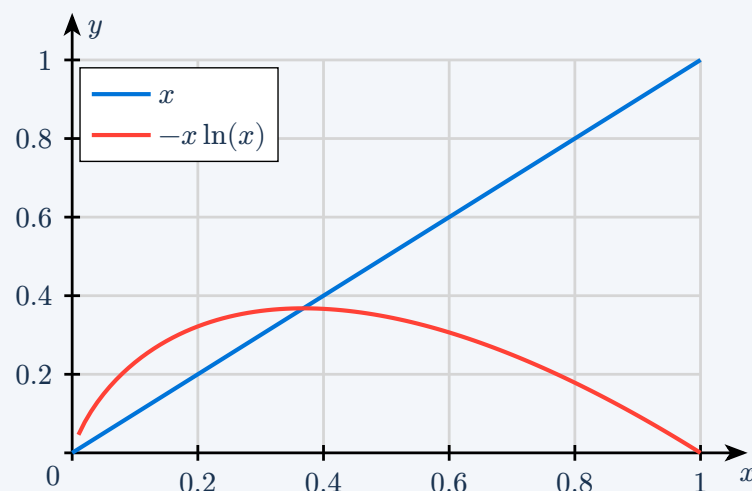
Curvature tells us how a function compares with its tangents, chords, and averages. Convex bends upward; concave bends downward.

1 Convex: above a tangent



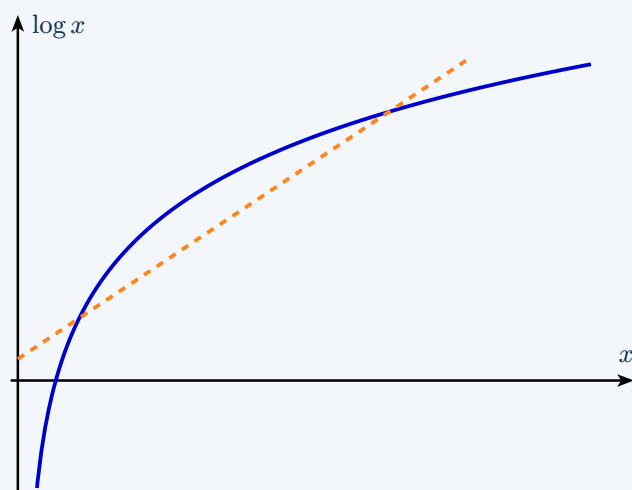
For $f(x) = x \ln x$ ($x > 0$), the tangent at $x = 1$ is $x - 1$. Thus $x \ln x \geq x - 1$, with equality at the contact point.

2 Concave: reverse the inequality



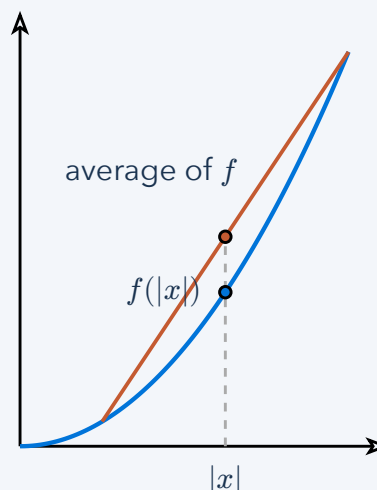
Negating $x \ln x$ reverses its curvature. A straight line is both convex and concave. Curvature statements apply on the specified domain.

3 The local picture for log



The concave logarithm lies below its tangent. To understand Jensen's inequality, compare a point on the curve with the chord between two sampled values.

4 Average first, or apply f first?



For convex f , $f(\sum_i w_i x_i) \leq \sum_i w_i f(x_i)$ when $w_i \geq 0$ and $\sum_i w_i = 1$. For concave f , reverse the sign.

Tangent = local slope; chord = interpolation between two points. Jensen compares the function of an average with the average of the function.